

Vanilla AE PSNR vs Latent Size

Autoresearch report · generated 2026-05-26

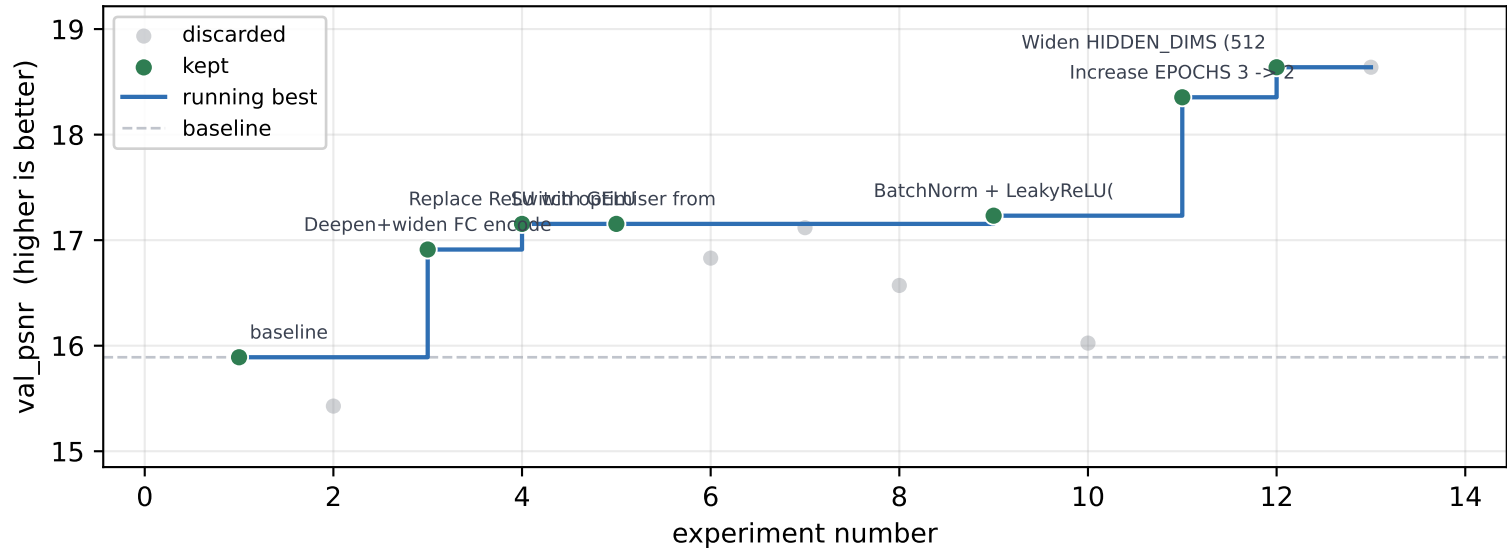
Experiment dates: 2026-05-26 Total run time: 6m 40s over 13 runs

Vanilla AE PSNR vs Latent Size — ran 13 experiments, kept 7 (discard 6, crash 0). Best val_psnr 18.6384 (baseline 15.8906, 2.7479 / +17.3%, higher is better).

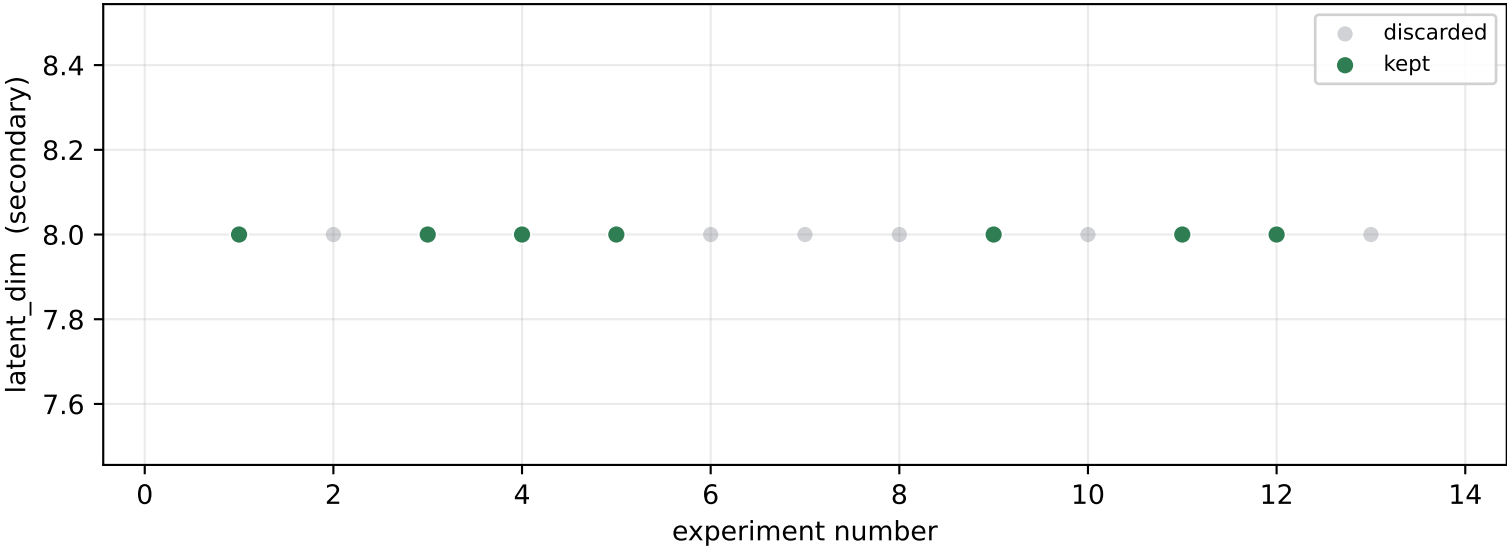
Goal

Iterate on a vanilla autoencoder so that reconstruction PSNR on a held-out validation split goes UP at a fixed latent (bottleneck) dimension. Primary success is higher `val\_psnr` than the baseline; `latent\_dim` is fixed in `prepare.py` (read-only) and is logged as a secondary metric purely for traceability — the loop must NOT change it.

Primary measure: val_psnr



Secondary: latent_dim



Kept experiments (improvements)

Exp #1	cdb0f7d	val_psnr 15.8906	Baseline
Baseline vanilla autoencoder: 784->128->32->128->784 fully-connected with ReLU, sigmoid output, MSE loss, Adam at lr 1e-3 for ~3 epochs. Establishes the reference val_psnr of 15.89 at the fixed latent_dim=8 that every later change is judged against.			
Exp #3	b7c3cfe	val_psnr 16.9112 (Δ +1.0207)	Paper
Deepened and widened the FC encoder/decoder to 784->512->256->64->latent (mirrored on the decoder side) with He initialisation, giving the network more capacity around the fixed bottleneck. Idea came from the paper 'Reducing Redundancy in the Bottleneck Representation of the Autoencoders' (arXiv:2202.04629). PSNR rose from 15.89 to 16.91 — the first real architectural win of the run.			
Exp #4	5f3935b	val_psnr 17.1541 (Δ +0.2429)	Paper
Swapped ReLU for GELU activations throughout the encoder and decoder so gradients flow more smoothly near zero. Motivated by 'SG-Blend: Learning an Interpolation Between Improved Swish and GELU' (arXiv:2505.23942), which highlights GELU's edge on reconstruction-style tasks. Lifted val_psnr from 16.91 to 17.15.			
Exp #5	e673751	val_psnr 17.1542 (Δ +0.0000)	Idea basket
Replaced Adam with AdamW (weight_decay=1e-4) to decouple L2 regularisation from the adaptive update, a curated suggestion from the idea basket. The effect was marginal numerically (17.1541 -> 17.1542) but the cleaner regularisation behaviour was preserved as a base for later changes. A small-but-keep gain on the run's higher-is-better metric.			
Exp #9	80d1d02	val_psnr 17.2319 (Δ +0.0778)	Idea basket
Inserted BatchNorm and switched activations to LeakyReLU(0.01) after each hidden Linear in both encoder and decoder, stabilising training of the deeper network. Drawn from the idea basket as a standard recipe for FC autoencoders. PSNR ticked up from 17.15 to 17.23.			
Exp #11	e78c4af	val_psnr 18.3538 (Δ +1.1219)	AI agent
Increased EPOCHS from 3 to 20 to actually use the 5-minute compute budget once the architecture had settled — an agent-led observation that the network was undertrained, not under-parameterised. This was the single biggest jump in the run: val_psnr 17.23 -> 18.35. Confirmed that further capacity gains would compound on top of longer training.			
Exp #12	239b9ed	val_psnr 18.6384 (Δ +0.2846)	AI agent
Widened HIDDEN_DIMS from (512,256,64) to (1024,512,128) to give the network more representational room outside the fixed bottleneck, keeping latent_dim=8. The agent's own reasoning after the longer-training win suggested capacity was the next bottleneck. New best PSNR of 18.64, up from 18.35.			